

supports. Engineers found themselves adapting existing technologies, merging them, on occasion, with traditional Indian practices, moulding them to suit each terrain and area.

Tunnelling through

Building tunnels was not new in India, but the ones constructed to accommodate trains were very different from those made for irrigation or military purposes. Unlike Europe, scholar Ian Derbyshire noted, tunnels were only required in a few places across India. These included the Western Ghats, the North-West Frontier Province, and the Himalayas. During 1850–70, tunnels were mainly built in the Western Ghats, on the Bhore and Thul Ghat inclines. There were 38 such tunnels of around 5,700 m aggregate length. Of these, two tunnels were the longest at about 490 m each.

The human cost

All of this came at a terrible cost. There was always the danger of severe injury or death, with cases of workers falling from heights while building bridges, drowning, accidents while cutting rock in the Ghats, or from blasting tunnels and cave-ins.

However, disease killed the most, cholera being the worst. The number of Indians that succumbed to this were far higher because of their terrible working and living conditions. One of the worst outbreaks was in 1860 in the Ghats where almost 25 per cent of Europeans died, with a far larger, undocumented number of deaths among the Indian labourers. “Jungle fever” or malaria also took its toll, often disabling workers.

Gradually, towards the end of the 19th century, worksites got better as did living conditions with sanitation and housing facilities, which helped in controlling mass outbreaks. Today, India has the fourth-largest railway network in the world, and a total track length of 121,407 km – a feat unimaginable without the human endeavour that started nearly 200 years ago.